Project Report: Retail Sales Analytics, Customer Segmentation, Churn Prediction and Sales Forecasting

Prepared By

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Project Overview

This project analyzes the Online Retail II dataset and develops a complete business analytics solution. The work includes data cleaning, exploratory data analysis (EDA), customer segmentation using RFM analysis and K-Means clustering, customer churn prediction using Random Forest, sales trend analysis, and demand forecasting using ARIMA, SARIMA, Prophet, and Exponential Smoothing models.

Objectives

• Clean and prepare retail transaction data.

• Analyze sales performance and customer behavior.

• Segment customers using RFM metrics.

• Predict customer churn using machine learning.

• Forecast future sales using time-series models.

• Generate business insights for decision-making.

Dataset Description

The project uses the Online Retail II dataset containing invoice details, customer IDs, quantities, product prices, invoice dates, and countries. Data from multiple sheets/years was merged into a single dataset for analysis.

Data Preprocessing

1. Combined datasets from multiple years.

2. Removed missing values and duplicate records.

3. Removed cancelled orders.

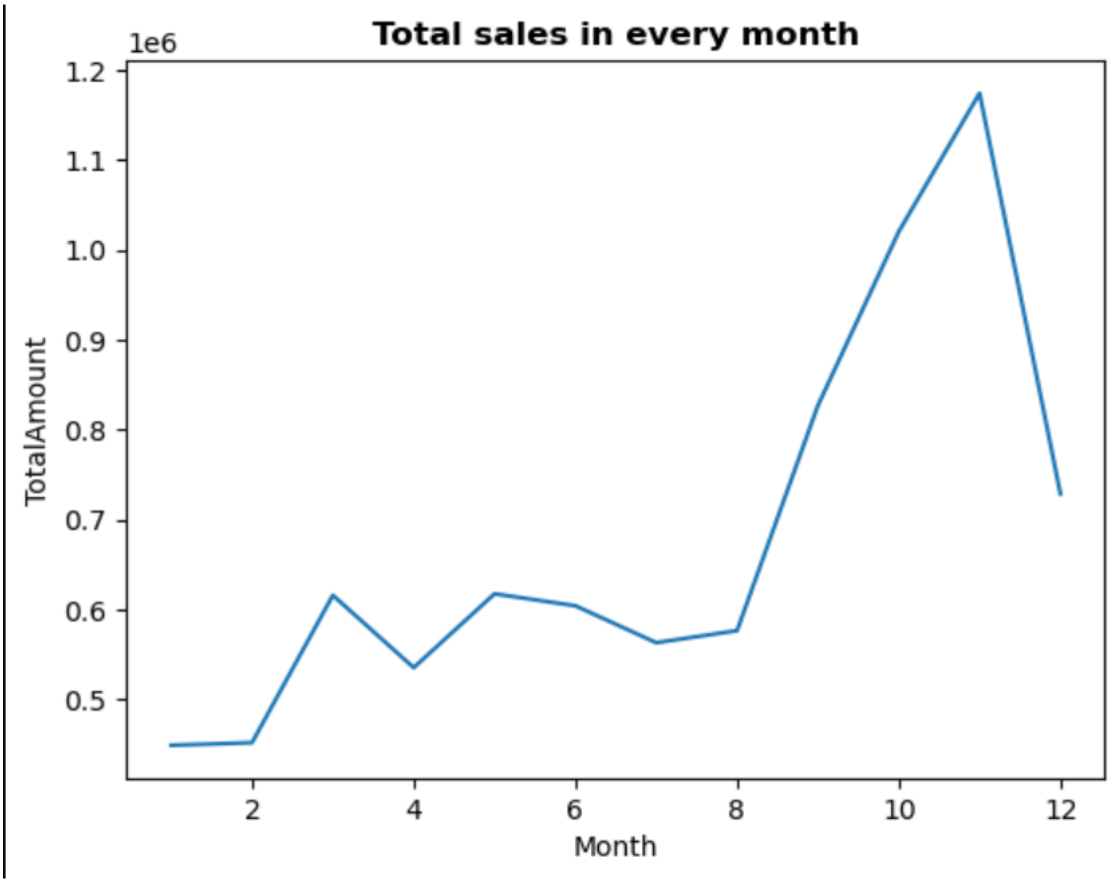
4. Filtered negative quantities and invalid prices.

5. Converted invoice dates into datetime format.

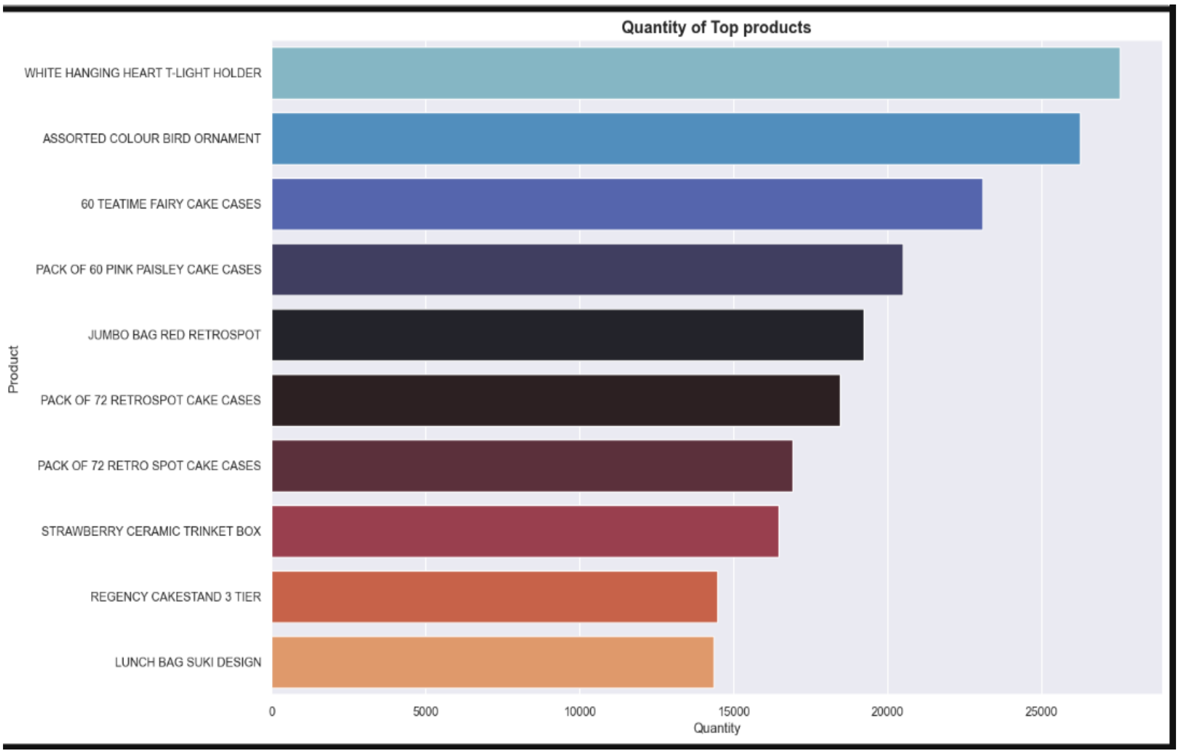
6. Created Total Amount = Quantity × Price.

Exploratory Data Analysis

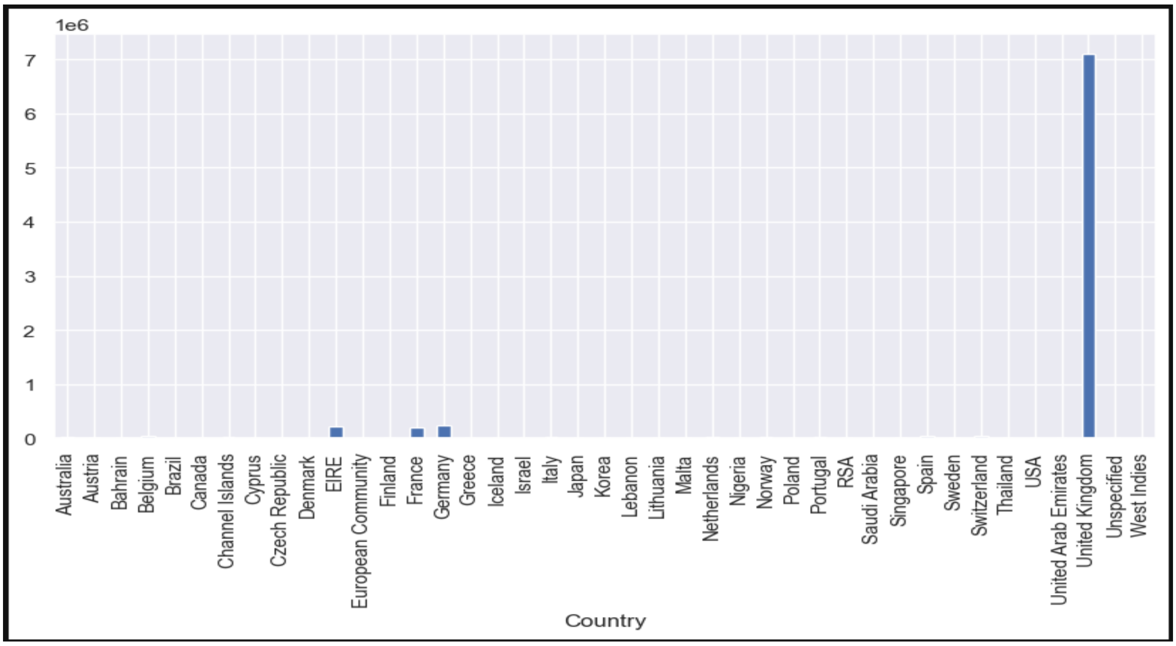
The analysis explored sales distribution, outliers, monthly sales trends, country-wise sales performance, and customer purchasing patterns. Visualizations such as histograms, box plots, line charts, bar charts, and pie charts were used to identify trends and business opportunities.



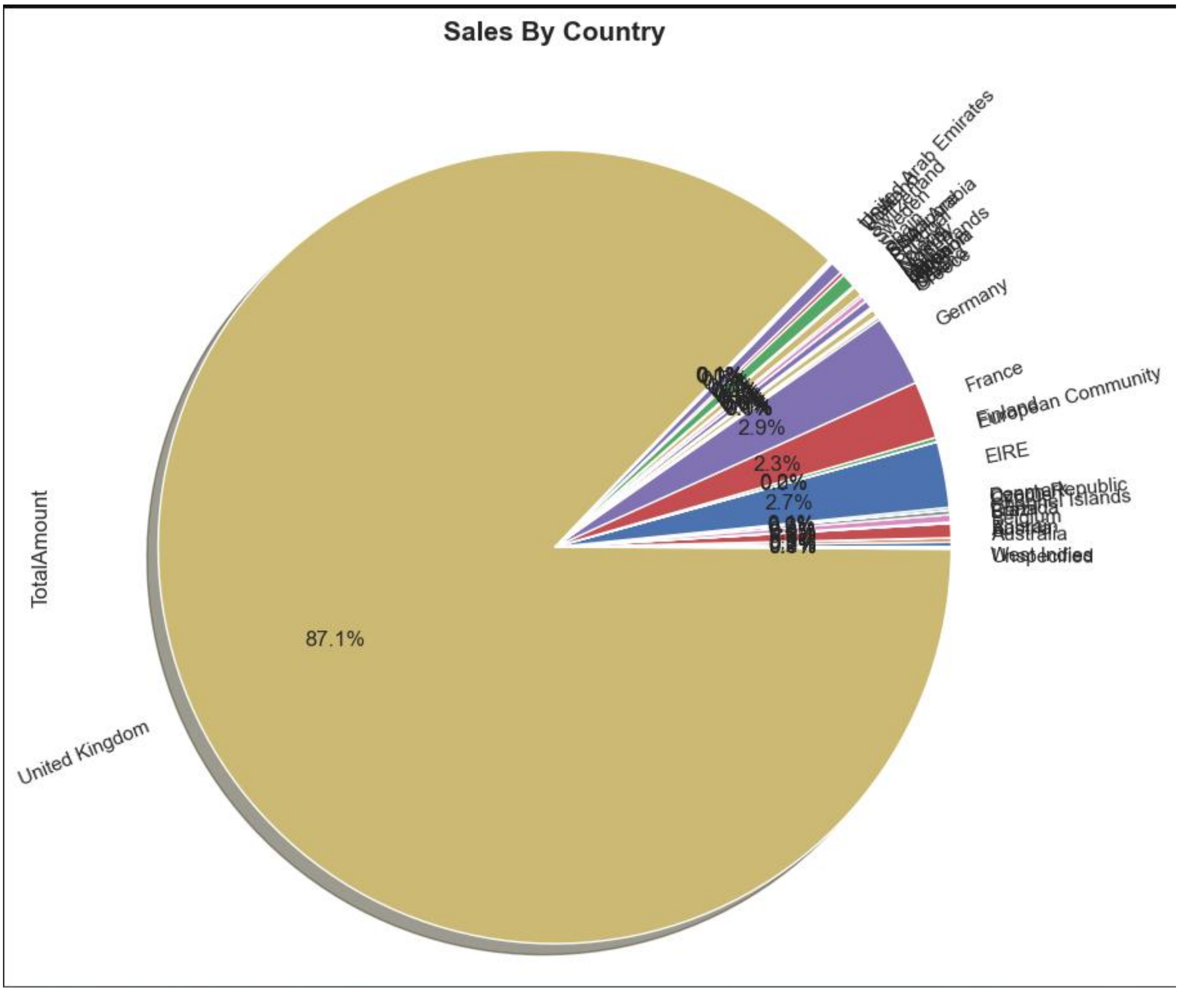
1. **Line Chart (** ***Total sales in every Month*****)**



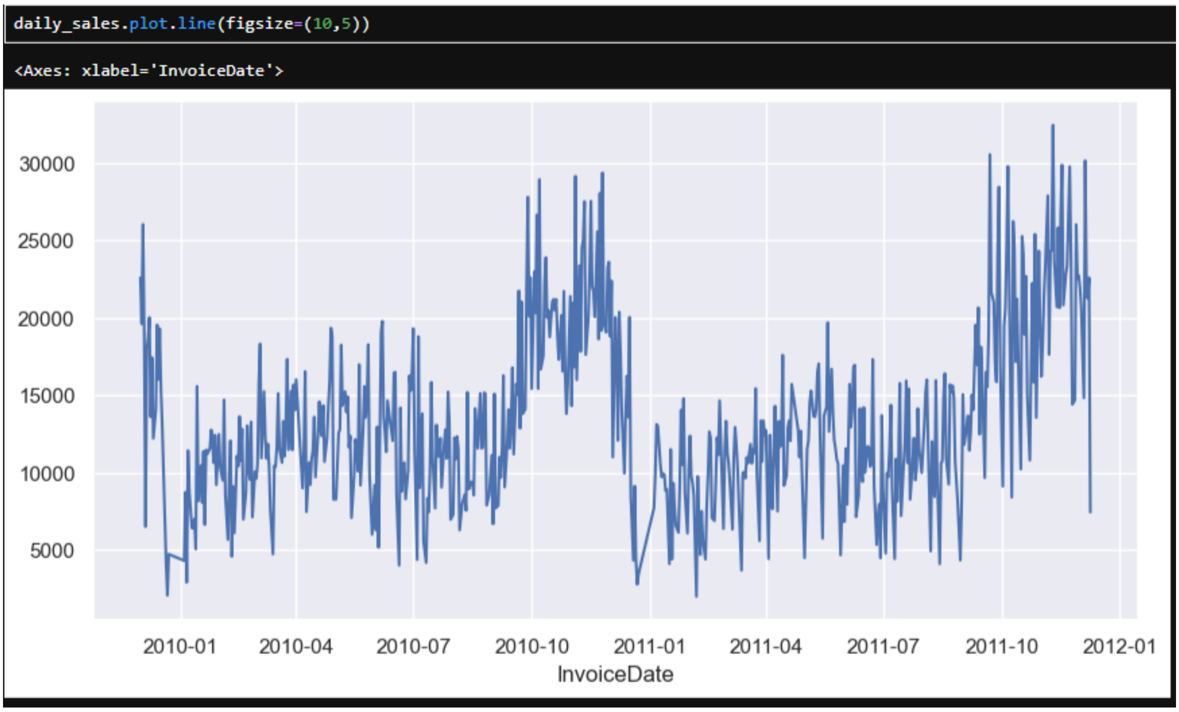
2. **Bar Chart (** ***Quantity By Product*** **)**



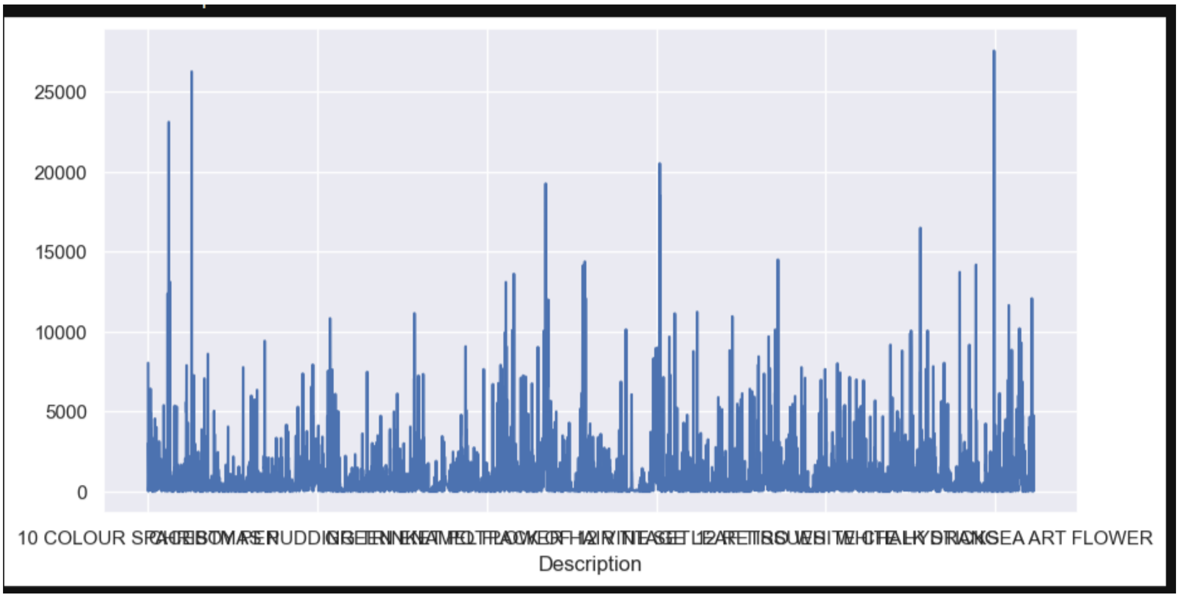
3. **Bar Chart (** *Country wise sales* **)**



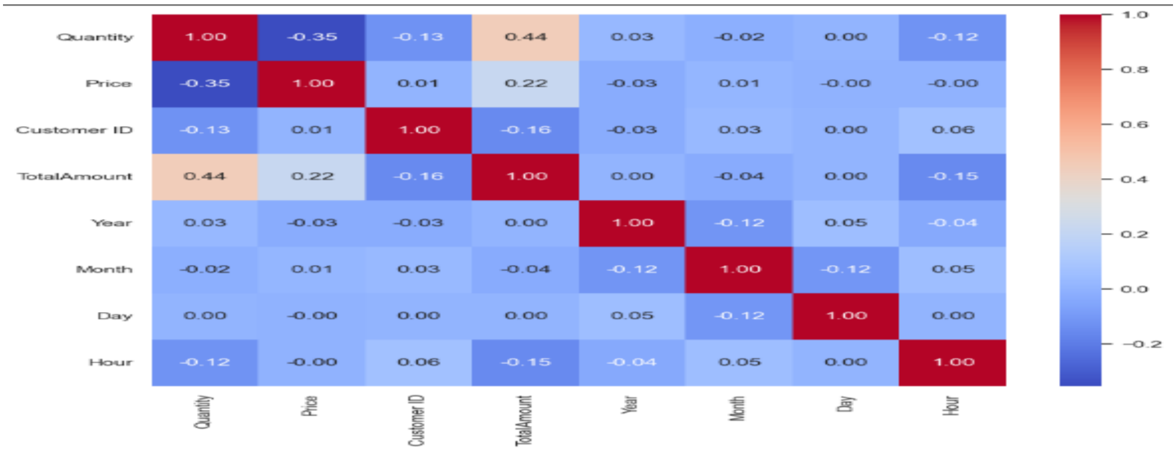
4. **Pie Chart (** *Country wise sales* **)**



5. **Line Chart (** *Daily wise Sales* **)**



6. **Bar Chart (** *Products wise Quantity* **)**



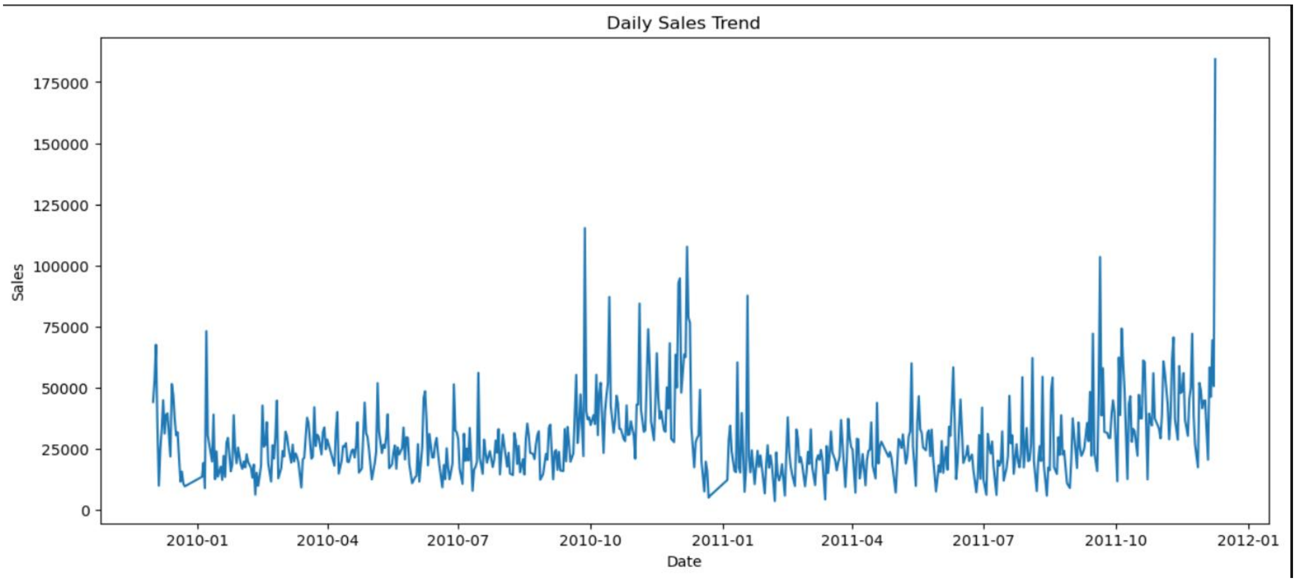
7. **Heatmap**

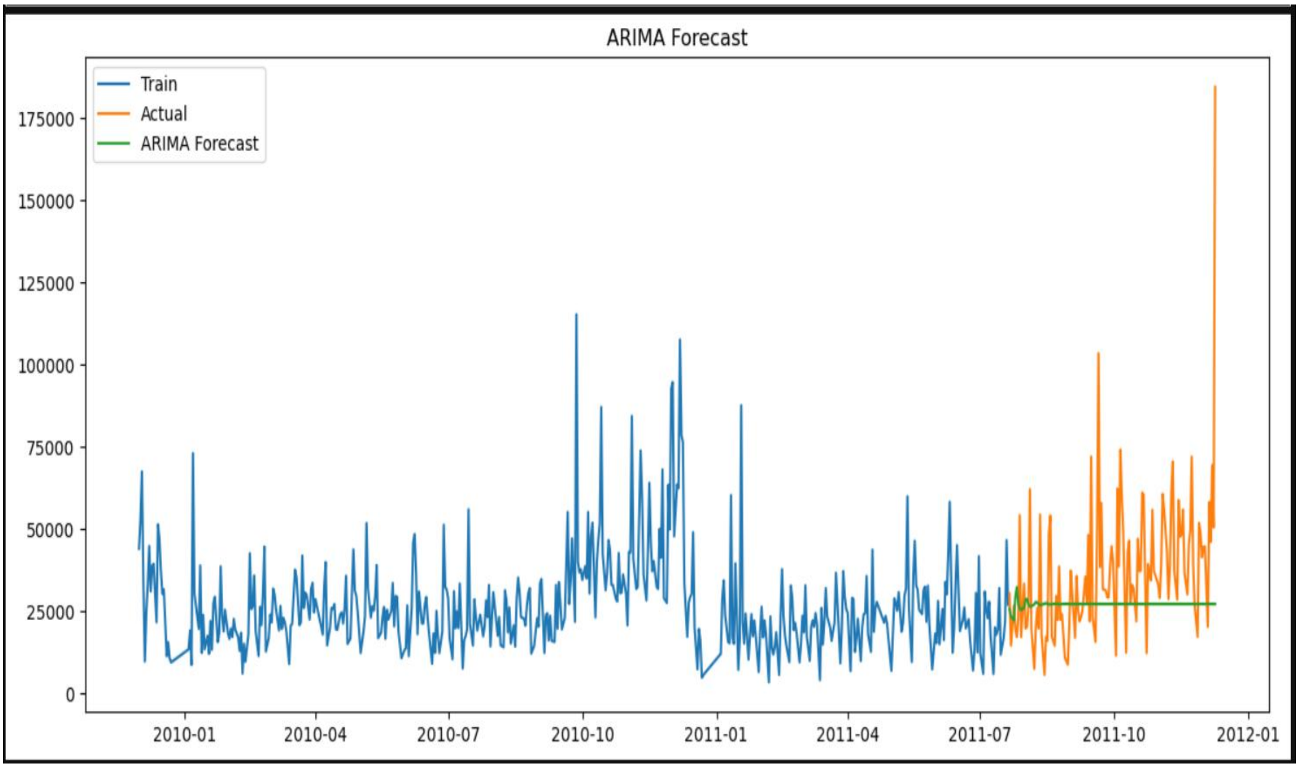
Customer Segmentation (RFM Analysis)

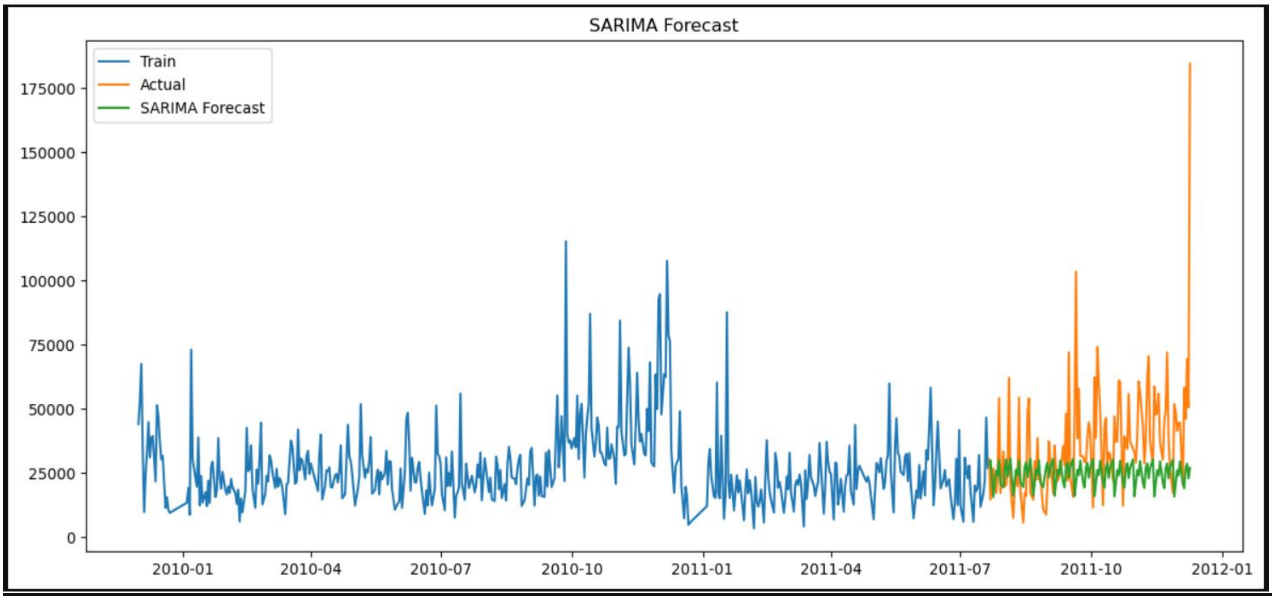
RFM analysis was performed using: • Recency – Days since last purchase. • Frequency – Number of unique purchases. • Monetary – Total customer spending. Standardization was applied and K-Means clustering (4 clusters) was used to segment customers into different business categories such as loyal customers, high-value customers, regular buyers, and at-risk customers.

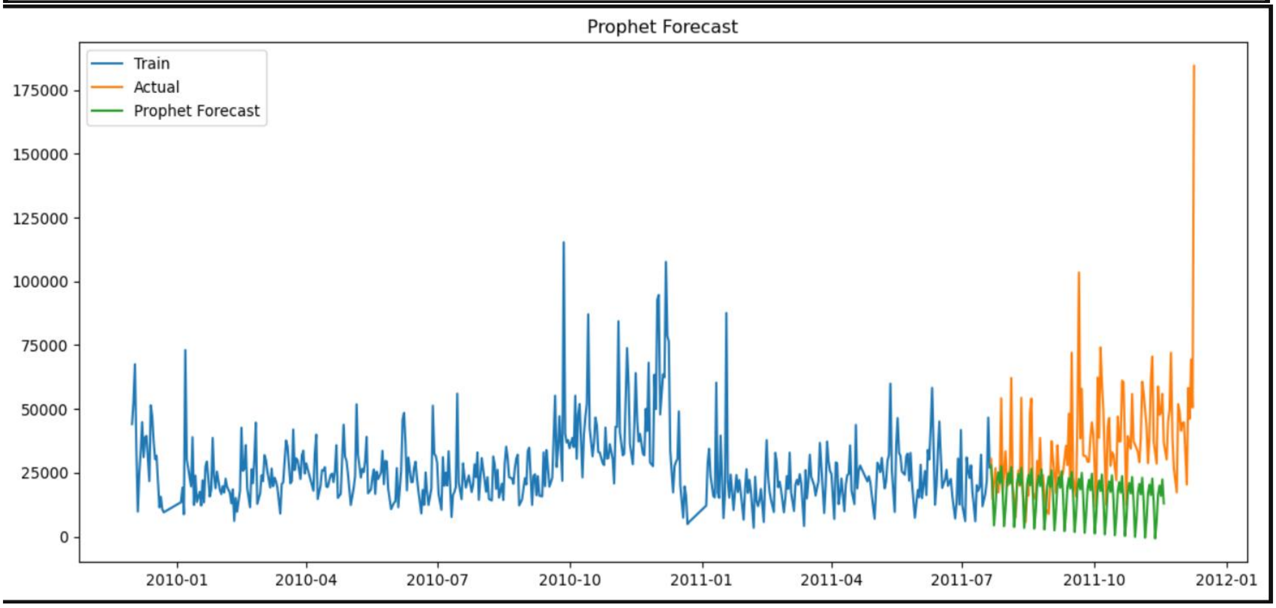
Sales Forecasting

Daily sales data was generated and forecasting models were applied: • Exponential Smoothing • ARIMA (5,1,2) • SARIMA (1,1,1) (1,1,1,12) • Prophet Forecasting performance was evaluated using MAE and RMSE metrics to compare prediction accuracy.









Churn Prediction

Customers with Recency greater than 90 days were marked as churned. Features used: • Recency • Frequency • Monetary A Random Forest Classifier was trained to predict customer churn and identify customers likely to stop purchasing.

Technology Stack

• Python

• Pandas

• NumPy

• Matplotlib

• Seaborn

• Scikit-Learn

• Stats models

• Prophet

• Power BI

Power BI Dashboard

A Power BI dashboard was developed to provide interactive business insights. The dashboard can display sales KPIs, regional performance, customer behavior, trends, and forecasting results for management reporting.

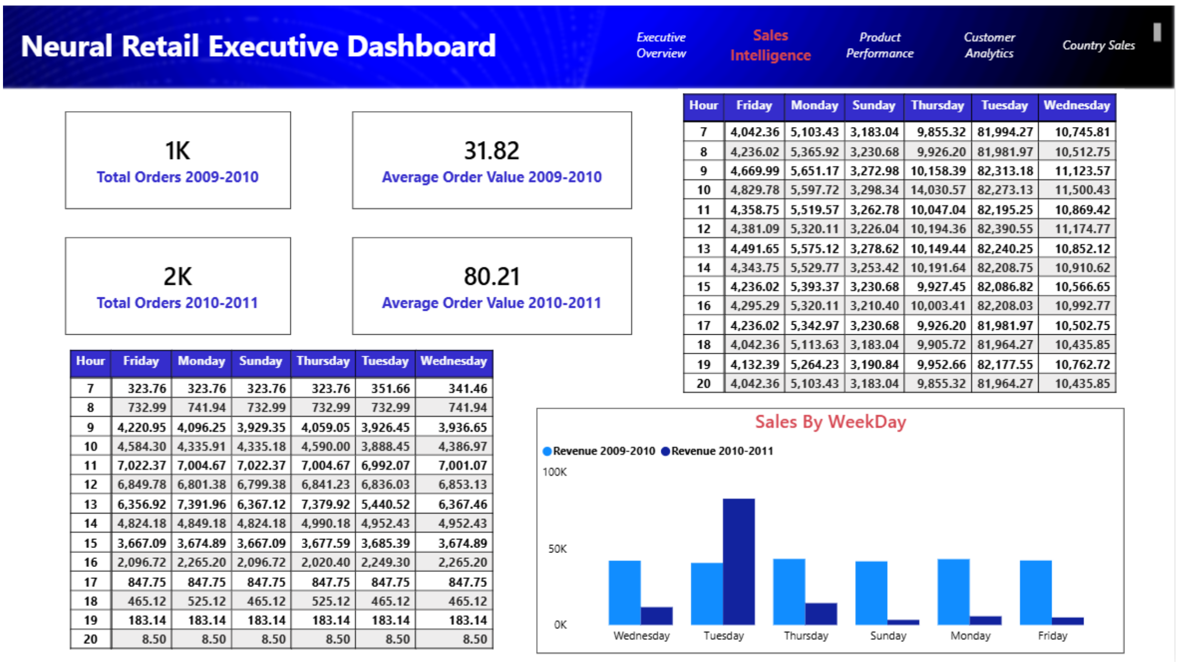
Dashboard Pages

1. Executive Summary

Provides a quick overview of key business KPIs such as Total Sales, Orders, Customers, and Profit.

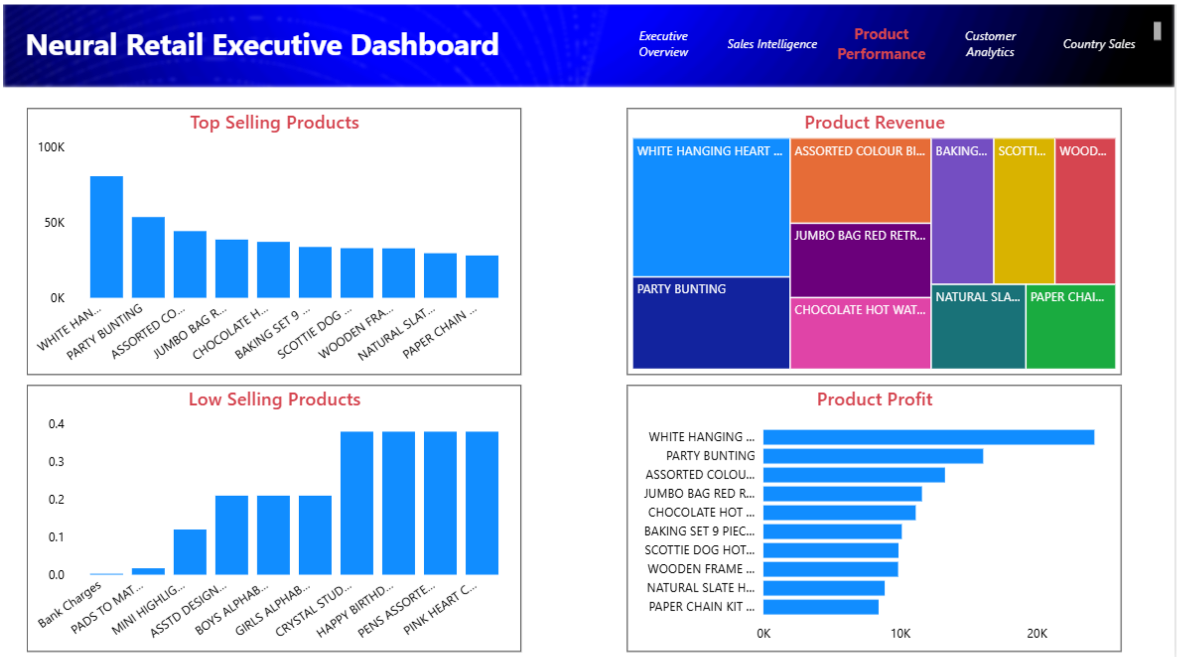
2. Sales Trend Analysis

Tracks sales performance over time and identifies growth patterns and seasonal trends.



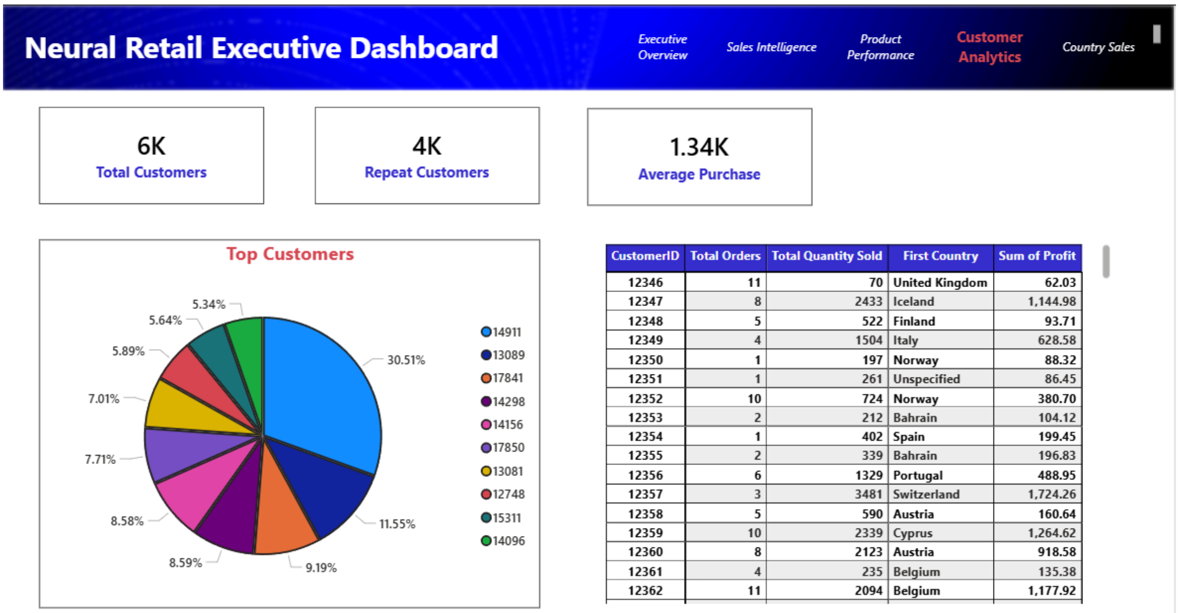
3. Product Performance

Highlights best-selling products, category-wise revenue, and product profitability.



4. Customer Analytics

Analyzes customer behavior, purchase frequency, and customer contribution to revenue.



5. Region & Country Sales

Displays geographical sales distribution and compares regional performance.

6. Monthly & Seasonal Analysis

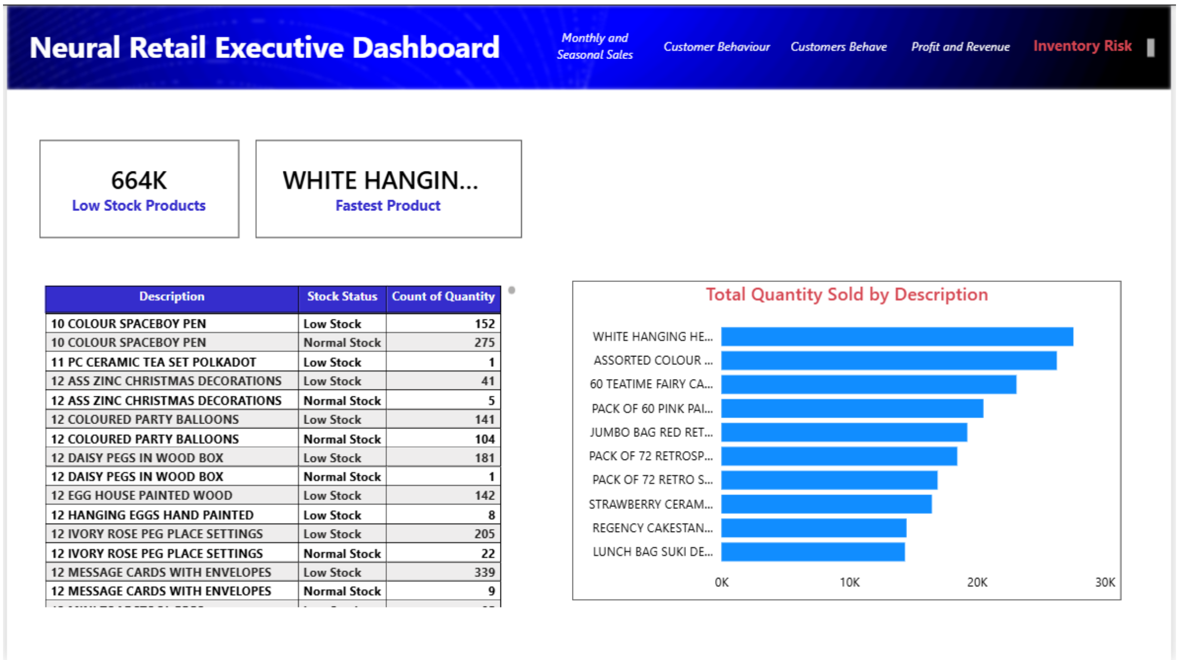
Identifies monthly sales patterns and seasonal demand fluctuations.

7. Customer Behaviour Dashboard

Uncover customer purchasing patterns and loyalty trends to drive retention, engagement, and revenue growth.

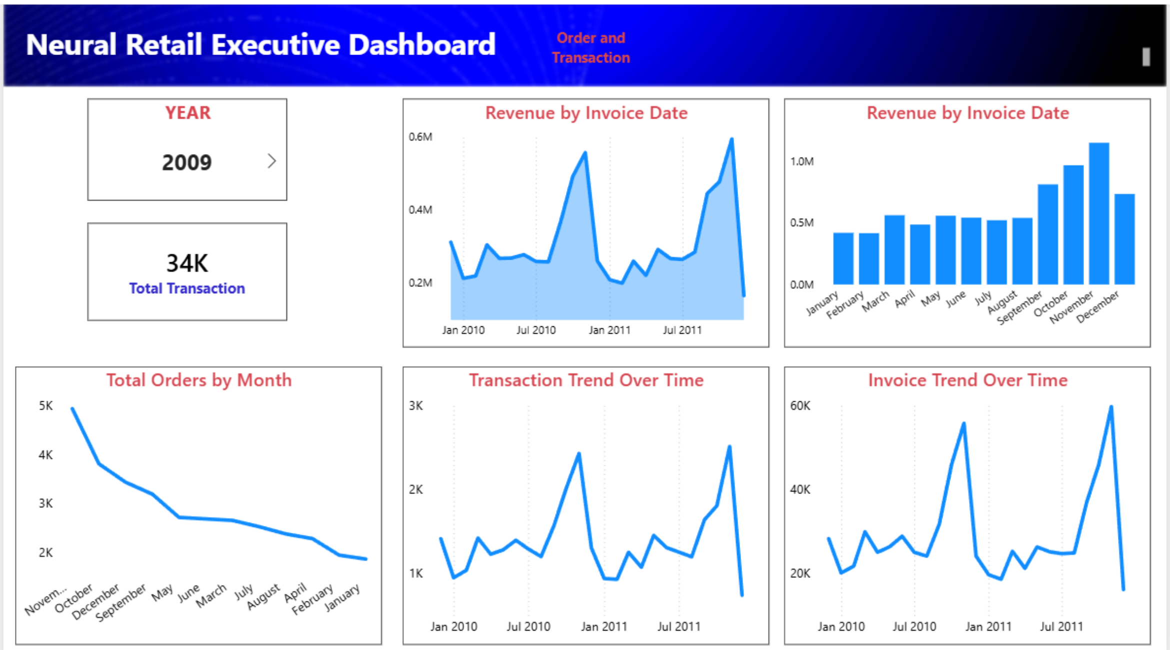
8. Inventory Analysis

Monitors stock levels and inventory performance to improve supply management.



9. Sales Forecasting

Predicts future sales trends using historical data for better planning.



10. Revenue & Profitability

Measures business profitability, revenue growth, and financial performance.

11. **Business Insights & Recommendations**

Summarizes key findings and provides actionable recommendations for business growth.

Business Benefits

• Better understanding of customer behavior.

• Identification of high-value customer groups.

• Early detection of churn risk.

• Improved inventory and sales planning.

• Data-driven business decision making.

Conclusion

The project successfully combines data analytics, machine learning, forecasting, and business intelligence techniques into a single retail analytics solution. Customer segmentation, churn prediction, and forecasting models provide actionable insights that can improve customer retention, revenue generation, and strategic planning.